



School of Computing

UNIVERSITI TEKNOLOGI MALAYSIA

**PROGRAMMING TECHNIQUE II
(SECJ1023)**

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FINAL PROJECT REPORT

SECTION 02

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PART 1: INTRODUCTION

1.1 Synopsis Project:

The synopsis of this project is to recommend the development of a Library System. This system includes Object-Oriented Programming (OOP) principles such as encapsulation, abstraction and association. Besides, this system will be designed to solve the problems faced by students and library administrators in managing the library records.

1.2 Objective of The Project

The objective of the project is the user can log in as either the students or the library administrators. Students can search the name of the books that they are interested in through this system and the details of the books such as the book's author and the availability status of the books will be displayed. Next, this system allows students to borrow books if the books are available and return books they have borrowed. Not only that, this system also enables the library administrators to view the fine status of the students and add new library books.

PART 2: SYSTEM ANALYSIS AND DESIGN (USE CASE AND CLASS DIAGRAM)

2.1 System Requirements

Use case diagram

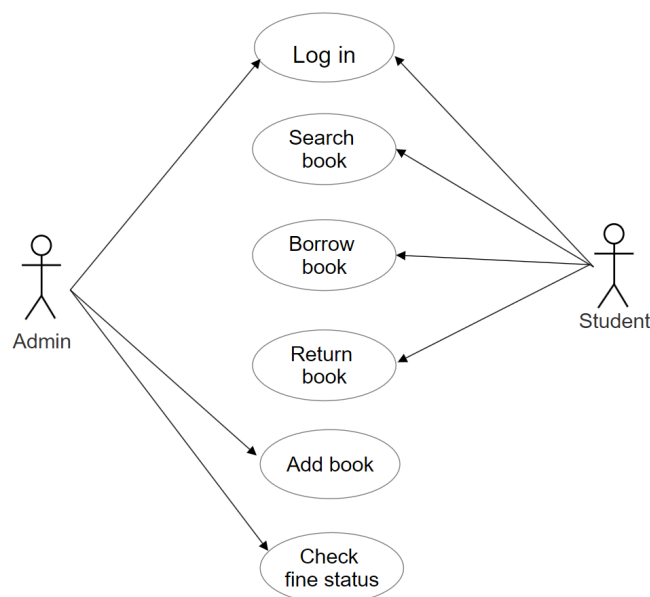


Figure 1 : Use Case Diagram for Library System

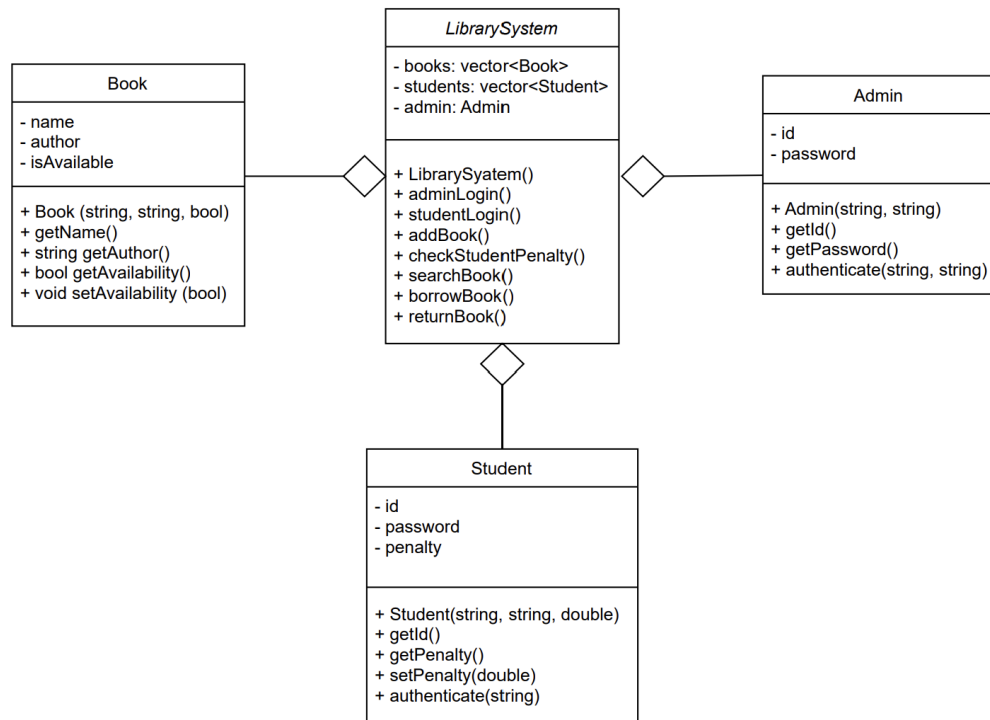
Detail Description for Each Use Cases

The system has 6 main use cases.

Use Case	Purpose
Log in	Enter ID and password for authentication purposes.
Search book	Search the name of the books and the details of the books such as the book's author and the availability status of the books will be displayed.
Borrow book	Provide choices for the user to borrow book if the book is available.
Return book	Enter the title of the book to return the book.
Add book	Key in the details of the books such as the book's name, book's author and the availability status of the books.
Check fine status	Enter student's ID to check the fine status of the student.

2.2 System Design

Class Diagram



PART 3: SYSTEM PROTOTYPE

The following figure shows an interface for UTM library system.

```
*****
LIBRARY SYSTEM
*****

= MAIN MENU =

SELECT OPTIONS
-----
1. ADMIN    2. STUDENT

(PRESS 3 TO EXIT)
```

Screen 1: Login page and main menu

Screen 1: The user must insert an integer value in the range 1-2 which enter 1 will enter admin interface or 2 to enter student interface. If the user enter other number, the system exit automatically.

```
1

Enter Admin ID: admin
Enter Admin Password: 123456

WELCOME, ADMIN! :) :)

Press 'p' to check student penalties, 'a' to add new books, or 'q' to log out:
```

Screen 2: Admin Login

Screen 2: After selecting admin option, admin should login with id and password for authentication purposes. Admin can enter 'p' to check student's fine status, 'a' to add new books or 'q' to quit and back to menu.

```
Press 'p' to check student penalties or 'a' to add new books: p
Please enter Student's ID to check his or her penalties: 123

There is a late fee of RM 25 for this student.
```

Screen 3: Checking Student Penalty System

Screen 3: Admin should enter student's ID to check for the penalties. The system will prompt student's totally late fee

```

Press 'p' to check student penalties or 'a' to add new books: a
Enter the name of the book: C++ Programming
Enter the author of the book: John
Is the book available for students to borrow? (y/n): y

///Book added successfully.///
Add more books? (y/n): |

```

Screen 4: Add New Books System

Screen 4: Admin should enter the name of the book ,the author and the availability of the book for students to borrow.The system will prompt a message that the book is added successfully. Admin can choose to enter “y” to add more books or enter“n” to stop

```

2
Enter Student ID: 123
Enter Student Password: 234567

WELCOME, STUDENT! ^_^ ^_^

Press 's' to search for books, 'b' to borrow books, 'r' to return books, or 'q' to log out:

```

Screen 5: Student Login System

Screen 5:This screen execute when user select 2 which is student interface.Student should login with id and password for authentication purposes.Student can enter ‘s’ to search for books ,enter ‘b’ to borrow books o,‘r’ to return books or ‘q’ to quit back to main menu.

```

Press 's' to search for books, 'b' to borrow books, or 'r' to return books: s
Please Enter the name of the book to search: Data Structure
** Book found **

Title           :Data Structure
Author          :Jane
Book Status     :Yes

```

Screen 6: Searching Book System

Screen 6: Student should enter book’s name and author to search a book.The system will prompt the details of the book if it is available such as title of the book,author and book status that shows that is it available to borrow .

```
Press 's' to search for books, 'b' to borrow books, or 'r' to return books: b
Please Enter the title of the book to borrow: Data Structure

// You have successfully borrowed the book: Data Structure //
```

Screen 7: Borrowing Book System

Screen 7: Student should enter book's name to borrow a book. The system will prompt a message that inform student that the book has been successfully borrowed if the book is available

```
Press 's' to search for books, 'b' to borrow books, or 'r' to return books: r
Please Enter the name of the book to return: Data Structure

// You have successfully returned the book: Data Structure //
```

Screen 8: Returning Book System

Screen 8: Student should enter book's name to return a book. The system will prompt a message that inform student that the book has been successfully returned

---- END OF DOCUMENTATION ----